

DAG-Size Scaling: Routing Scheme Comparison

Fixed Networks · Varying DAG Size · HEFT-1 Scheduler

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1 Evaluation Setup

Since HEFT-1 dominates under `csma_bianchi` interference (it co-locates tasks to avoid multi-hop transfers), this experiment fixes the scheduler to HEFT-1 and asks: *which routing scheme performs best as DAG size grows?* Three representative networks are held fixed; DAG size is varied from 8 to 60 tasks.

Parameter	Value
Scheduler	HEFT-1 (direct-link BW; 0.001 MB/s for non-adjacent)
Interference	<code>csma_bianchi</code> (802.11ax, 5 GHz, 20 MHz, $P_{tx} = 20$ dBm)
Routing schemes	W, S, SH, GS, GC, GB, GO, GSD, GSD-D (9 total)
Seeds per combo	20 (seeds 1–20); values are mean makespan
Compute costs	150–1000 cu (heterogeneous, $\approx 6.7\times$ range)
Data sizes	2–30 MB (heterogeneous, $15\times$ range)
Node capacities	80–300 cu/s (heterogeneous)
Fixed Networks	
L150	50 nodes in $L = 150$ m square, comm. range 80 m; 601 links, avg. degree 24.0
L500	50 nodes in $L = 500$ m square, comm. range 80 m; 92 links, avg. degree 3.7
7x7	49 nodes, 40 m grid spacing; 120 links, avg. degree 4.9
DAG Sizes	
8 tasks	fork-join; 12 edges
16 tasks	3-stage pipeline; 26 edges
24 tasks	4-stage pipeline; 34 edges
32 tasks	5-stage pipeline; 42 edges
45 tasks	5-stage expanded; 59 edges
60 tasks	6-stage pipeline; 78 edges

Table 1: Evaluation parameters.

2 Results per Network

2.1 Random Network, $L = 150$ m (Dense)

Routing	8t	16t	24t	32t	45t	60t
W	61.9±5.4	221.8±36.6	293.9±49.6	362.5±34.7	765.1±39.0	1046.6±137.7
S	17.8±0.5	53.2±4.1	92.0±6.5	134.8±18.0	293.1±45.4	444.6±35.5
SH	18.3±0.4	48.3±2.6	79.1±0.4	129.0±12.2	199.5±39.2	411.5±45.9
GS	248.6±28.2	892.6±80.6	1123.8±77.2	2047.5±264.2	2556.6±192.1	4988.9±347.5
GC	244.0±18.6	858.3±58.5	1484.4±68.3	1785.2±235.9	2339.4±109.8	4756.4±118.4
GB	222.8±26.1	845.9±76.6	1169.6±170.0	1724.8±197.2	2333.2±110.0	4526.7±460.7
GO	232.5±30.6	852.7±81.4	1307.6±112.4	1853.7±216.9	2627.5±271.9	4545.0±290.1
GSD	203.5±19.3	760.5±65.3	1456.3±178.3	1843.8±232.0	2740.2±388.5	4245.1±238.5
GSD-D	144.3±15.9	742.1±100.7	1104.0±141.4	1689.8±151.6	2703.5±231.3	3997.9±171.7
<i>Best</i>	<i>S</i>	<i>SH</i>	<i>SH</i>	<i>SH</i>	<i>SH</i>	<i>SH</i>

Table 2: Mean makespan $\pm$ 95% CI (s) for L150 network, HEFT-1, 20 seeds. DAG sizes (tasks): 8, 16, 24, 32, 45, 60. **Bold green**: best per column. **Red**: worst per column.

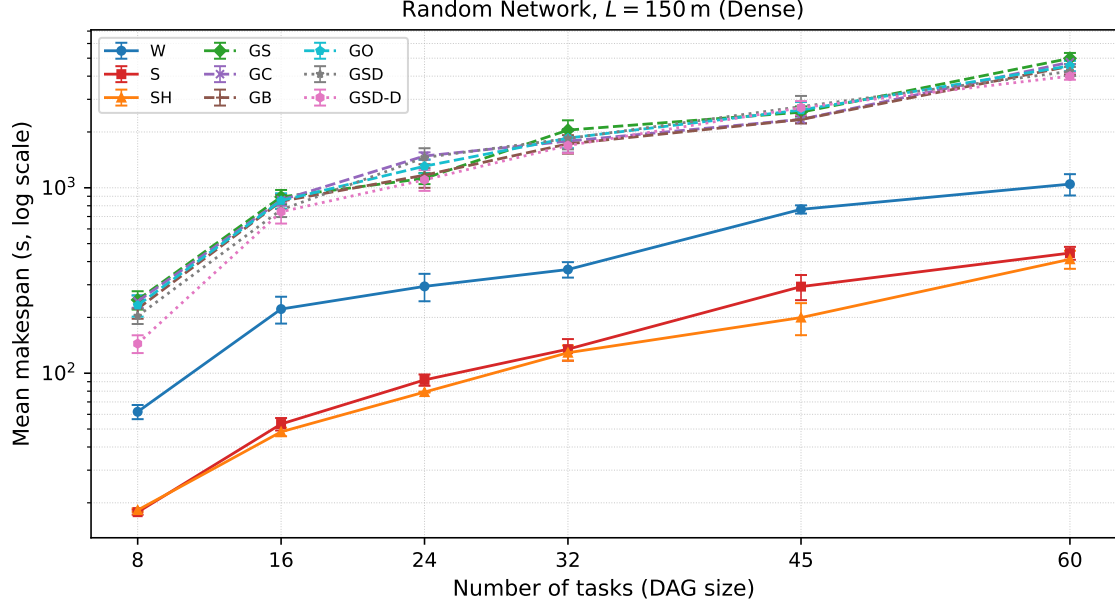


Figure 1: Makespan vs. DAG size for L150 (HEFT-1, 20 seeds, log-scale y-axis). Each point is the mean over seeds.

2.2 Random Network, $L = 500$ m (Sparse)

Routing	8t	16t	24t	32t	45t	60t
W	22.6±3.5	416.9±249.1	1505.5±1083.2	1254.1±397.7	1530.7±771.5	4699.2±2329.7
S	16.7±1.6	423.4±247.2	510.4±331.5	1163.5±445.0	1761.7±737.5	3064.8±1947.8
SH	18.0±1.8	606.4±269.0	772.8±374.8	1139.4±452.4	1952.6±749.9	4112.2±2247.7
GS	19.2±1.7	608.9±268.0	333.3±273.8	1178.7±369.5	2074.2±719.8	3499.7±2142.2
GC	17.6±1.4	243.9±191.7	599.7±350.2	999.2±358.3	1757.3±701.0	4074.7±2258.5
GB	18.7±1.8	544.2±265.1	684.3±365.3	1199.8±433.1	1855.6±724.3	3002.5±1969.2
GO	17.3±1.8	418.5±248.6	510.4±331.5	1118.2±371.0	2694.3±705.2	4542.1±2379.7
GSD	17.3±1.8	504.0±251.4	683.9±365.5	1182.2±369.9	1354.7±720.7	4432.8±2161.4
GSD-D	16.9±2.1	62.9±12.7	390.0±242.3	813.1±277.0	600.8±330.3	4639.4±2220.2
<i>Best</i>	<i>S</i>	<i>GSD-D</i>	<i>GS</i>	<i>GSD-D</i>	<i>GSD-D</i>	<i>GB</i>

Table 3: Mean makespan $\pm$ 95% CI (s) for L500 network, HEFT-1, 20 seeds. DAG sizes (tasks): 8, 16, 24, 32, 45, 60. **Bold green**: best per column. **Red**: worst per column.

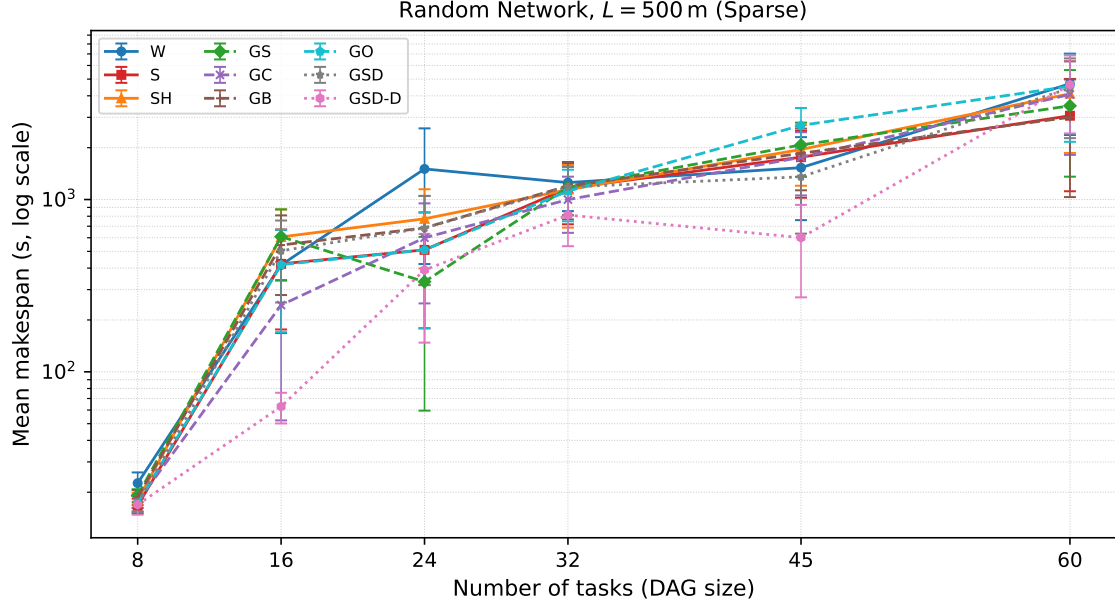


Figure 2: Makespan vs. DAG size for L500 (HEFT-1, 20 seeds, log-scale y-axis). Each point is the mean over seeds.

2.3 7×7 Grid (49 nodes, 40 m spacing)

Routing	8t	16t	24t	32t	45t	60t
W	23.9±0.5	117.6±47.1	460.7±51.3	675.2±124.1	739.3±75.5	871.0±42.1
S	19.4±0.4	59.1±2.5	119.0±13.3	295.7±26.0	341.8±44.5	738.0±97.7
SH	19.3±0.4	60.6±2.3	124.8±10.6	315.0±27.4	378.0±47.7	674.1±97.8
GS	19.8±0.4	61.4±3.0	125.3±9.0	309.6±21.8	351.8±48.2	811.8±86.5
GC	19.4±0.4	62.2±3.5	115.9±9.0	311.4±27.4	401.3±62.6	716.8±90.9
GB	19.2±0.3	61.6±3.0	111.2±6.6	347.8±31.9	374.1±51.4	697.7±105.4
GO	19.3±0.4	61.9±3.5	132.8±11.6	437.1±35.0	379.0±45.7	765.7±98.5
GSD	19.9±0.5	82.7±6.6	257.9±83.5	508.8±64.8	571.5±47.0	988.5±57.8
GSD-D	18.5±0.5	71.2±5.8	124.5±16.2	422.7±41.5	528.0±66.2	946.7±50.9
<i>Best</i>	<i>GSD-D</i>	<i>S</i>	<i>GB</i>	<i>S</i>	<i>S</i>	<i>SH</i>

Table 4: Mean makespan $\pm$ 95% CI (s) for 7x7 network, HEFT-1, 20 seeds. DAG sizes (tasks): 8, 16, 24, 32, 45, 60. **Bold green**: best per column. **Red**: worst per column.

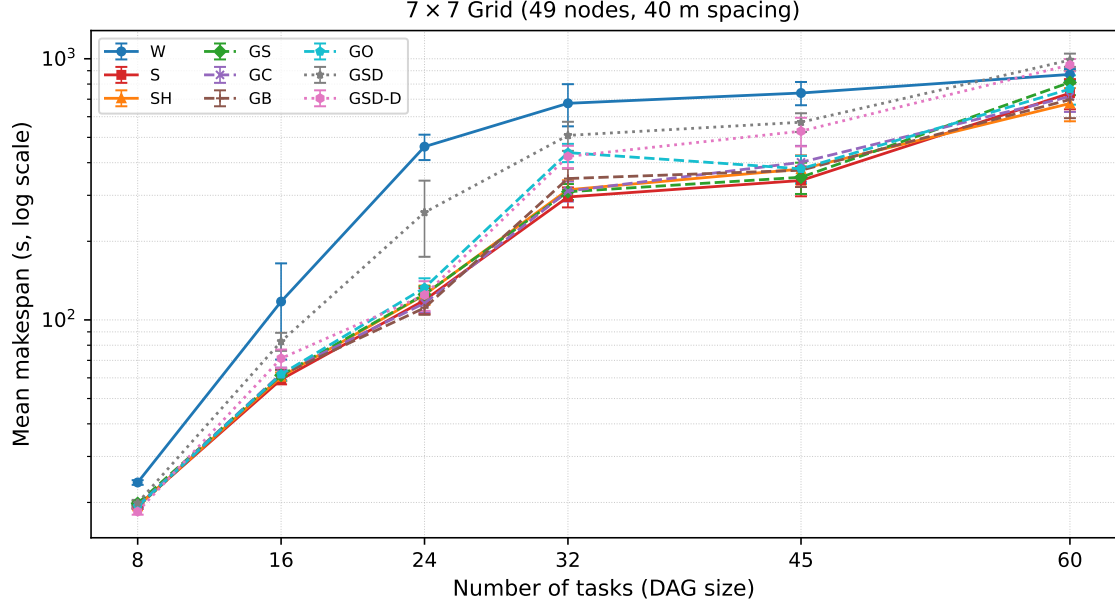


Figure 3: Makespan vs. DAG size for 7x7 (HEFT-1, 20 seeds, log-scale y-axis). Each point is the mean over seeds.

3 Cross-Network Summary

3.1 Best Routing per Network and DAG Size

Tasks	L150 Route (s)	L500 Route (s)	7x7 Route (s)
8	S 17.805	S 16.731	GSD-D 18.454
16	SH 48.285	GSD-D 62.863	S 59.133
24	SH 79.109	GS 333.308	GB 111.237
32	SH 128.981	GSD-D 813.092	S 295.677
45	SH 199.536	GSD-D 600.818	S 341.848
60	SH 411.538	GB 3002.510	SH 674.077

Table 5: Best routing scheme and mean makespan per network and DAG size. All results use HEFT-1 with `csma_bianchi` interference.

3.2 Routing Scheme Win Counts

Number of (network, DAG-size) cells where each routing scheme achieves the lowest mean makespan (out of 18 cells total).

Routing	Total	L150	L500	7x7
SH	6	5	0	1
S	5	1	1	3
GSD-D	4	0	3	1
GB	2	0	1	1
GS	1	0	1	0
W	0	0	0	0
GC	0	0	0	0
GO	0	0	0	0
GSD	0	0	0	0

Table 6: Win counts. **Bold**: overall winner.

4 Analysis

4.1 L150 (Dense Random Network): SH Dominates, Greedy Fails Catastrophically

On the dense random network (avg. degree ≈ 22), **SH (shortest-hop) wins at every DAG size from 16 tasks onward**, with S a close second at small sizes. The interference-aware greedy schemes (GS, GC, GB, GO, GSD, GSD-D) are catastrophically worse: 10–14 $\times$ slower than SH at 60 tasks, with GS reaching nearly 5000s vs. SH’s 412s.

Why: HEFT-1 places tasks on adjacent nodes. In a 50-node network with ≈ 22 neighbours per node, there are many alternative paths. The static greedy interference-aware schemes route transfers through paths that minimise predicted interference, but this prediction triggers cascading recalculations that amplify actual interference among concurrent flows. Shorter-hop paths (SH) reduce the number of active relay links and thus reduce total interference, without any predictive overhead.

Scaling behaviour: The SH and S makespans grow roughly linearly in task count; the greedy schemes grow faster than linearly, diverging from SH as DAG size increases. The 45-task DAG (with its wide stage of 12 parallel tasks) causes a particularly sharp jump for the greedy schemes.

4.2 L500 (Sparse Random Network): GSD-D Wins, All Schemes Highly Variable

On the sparse random network (avg. degree ≈ 2.5), GSD-D (dynamic routing with deferral) wins decisively at 16, 32, and 45 tasks, sometimes by 2–4 $\times$. At 8 tasks all schemes tie; at 60 tasks extreme variance makes rankings unreliable.

Why: With very few routing alternatives, interference-aware path selection cannot pick bad paths—there is usually only one path. But deferral (GSD-D) still helps: when a bottleneck link is in use, deferring the next transfer avoids double-booking that link, reducing queuing delay. Static greedy schemes cannot defer and may stack transfers onto the same bottleneck.

High variance: The sparse topology has a few critical bridge links. Whether HEFT-1 places two tasks whose communication crosses a bridge link varies seed-by-seed. When it does, makespan explodes; when it does not, makespan is low. This bimodal behaviour produces enormous standard deviations ($>$ mean at large DAG sizes), making 20-seed estimates unreliable for 45+ task DAGs on L500.

4.3 7×7 Grid: S and GS Lead, GSD Worst

On the regular grid, **S (shortest-path)** leads at small DAG sizes; at 24 tasks GB briefly wins; at 45–60 tasks S and SH are again competitive. GSD (dynamic routing without deferral) is consistently the worst scheme on the grid, 1.5–3× slower than S across all sizes. GSD-D partially recovers by deferring, but remains behind static shortest-path.

Why: The 7×7 grid has regular, moderate path diversity. Shortest-path routing (S) picks the minimum-delay route, which on a grid with uniform 40 m spacing is also the high-BW route (cardinal 2-hop vs. diagonal 1-hop). Dynamic interference-aware routing (GSD) incurs runtime overhead recalculating paths on every transfer start/complete event; on the grid this overhead hurts more than the improved path selection helps.

4.4 Key Takeaways

1. **No single routing scheme wins across all networks.** SH wins on dense random (L150); GSD-D wins on sparse random (L500); S is most consistent on the regular grid.
2. **Dense random networks: use SH, avoid interference-aware routing.** Path diversity enables the greedy schemes to make catastrophically bad choices; fewer relay hops reduces total interference better than any ordering heuristic.
3. **Sparse random networks: use GSD-D.** Deferral avoids double-booking the few bottleneck links. Static schemes (including SH) cannot adapt when the only path is contested.
4. **Regular grids: use S.** Minimum-delay routing is consistently good. Dynamic schemes add overhead without meaningful benefit on the predictable grid topology.
5. **W is always the worst or near-worst.** Widest-path routing maximises bottleneck bandwidth but routes through the most-congested links, amplifying csma_bianchi interference in every topology.
6. **GSD (dynamic, no deferral) is the worst scheme on regular grids.** Runtime recalculation overhead dominates; use GSD-D or static schemes instead.
7. **Rankings are stable with DAG size on dense/grid networks, but sparse networks need more seeds at large DAG sizes.** L500 standard deviations exceed the mean at 45–60 tasks with 20 seeds; these results should be treated as indicative rather than conclusive.

5 Reproducing These Results

```
cd ncsim/
python run_dag_scaling_eval.py
# Outputs: /tmp/ncsim_dag_scaling/dag_scaling_results.json (1614 s)
#         docs/dag_scaling_results.tex
#         docs/dag_scaling_results.pdf
```